

Software Requirements Specification (SRS)

Online Exam Portal

Project: Online Exam Portal

Team: Team 15

Authors: Chiranthan V (PES2UG24CS138) · Dwijesh Krishna Ghattamaraju (PES2UG24CS163) · Harkirat Singh (PES2UG24CS180)

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Revision history

Version	Date	Author	Change summary	Approval
1.0	05-09-2026	Team 15	Merged supporting UML diagrams, system architecture and database design into a single self-contained SRS for final submission	Final

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This document is the complete, self-contained Software Requirements Specification (SRS) for the Online Exam Portal, a web-based system that lets an Admin (faculty/exam coordinator) build a reusable MCQ question bank, assemble and schedule exams from it, and lets Students attempt those exams under a server-enforced timer with automatic evaluation and reporting. It defines functional and non-functional requirements, interfaces, security requirements, system models/diagrams, database design, and verification criteria for the development team and course evaluators. No separate report or appendix is required to understand the system.

1.2 Scope

Covers Admin and Student web interactions: authentication, question-bank management, exam creation/scheduling, timed exam attempts with auto-save, server-side auto-evaluation, result/analytics reporting, bulk student import, and basic tab-switch proctoring signals. Excludes descriptive/subjective-question grading, live webcam proctoring, plagiarism detection, and integration with external university ERP/payment systems.

1.3 Audience

Developers, QA testers (Team 15), course instructor/evaluators, and prospective student/admin users participating in a pilot/demo.

1.4 Definitions

MCQ (Multiple Choice Question), RBAC (Role-Based Access Control), JWT (JSON Web Token), NFR (Non-Functional Requirement), FR (Functional Requirement), RTM (Requirements Traceability Matrix), TLS (Transport Layer Security), CSV (Comma-Separated Values), WCAG (Web Content Accessibility Guidelines), UML (Unified Modeling Language), ERD (Entity-Relationship Diagram), PK/FK (Primary Key/Foreign Key).

2. Overall description

2.1 Product perspective

The Online Exam Portal is a standalone, self-hosted three-tier web application (client, application server, database) used within a single department/institution. It is not a component of a larger commercial LMS; it interacts only with an SMTP service for email notifications. The full system architecture is presented as Figure 6 in Section 7.6.

2.2 Major product functions (detailed)

- Admin/Student authentication with RBAC
- Question bank management (create/edit/delete, tagging)
- Exam creation, question selection (manual/random) and scheduling
- Batch/section-based exam assignment and student bulk import
- Timed exam attempt with auto-save and server-enforced auto-submit
- Automatic server-side evaluation with negative marking
- Result and class/question-wise analytics reporting (CSV/PDF export)
- Tab-switch/focus-loss proctoring-lite logging and admin review
- Email notifications for exam scheduling and result declaration

2.3 User roles and characteristics (expanded)

- Admin (Faculty/Exam Coordinator): comfortable with spreadsheets/CSV; needs an efficient question-bank workflow and clear analytics; trusted role, not adversarial.

- Student: basic computer literacy; expects a clear timer, no ambiguity about time remaining, and instant, fair results.

2.4 Operating environment

Modern evergreen browsers (Chrome, Firefox, Edge) on desktop and mobile; server deployed on a Linux VM/container (Docker) with PostgreSQL; institutional or campus Wi-Fi/LAN network.

2.5 Constraints

Must be deliverable within one academic semester by a 3-person student team; modest infrastructure budget (single VM or free-tier cloud instance for the demo); browser-based proctoring cannot detect a second physical device — stated as an explicit, accepted limitation.

3. External interface requirements

3.1 User interfaces

Primary UI: responsive web pages (desktop/mobile browser) with a persistent, always-visible countdown timer during an attempt; high-contrast mode and keyboard navigation for accessibility.

3.2 Hardware interfaces

None beyond a standard client device (desktop/laptop/tablet) with keyboard/mouse or touchscreen and a working network interface. No specialized hardware is required.

3.3 Software interfaces

- Application API (Django REST Framework or Express.js) over HTTPS/JSON for all client-server communication
- SMTP service (e.g., SendGrid/institutional mail server) for notification emails
- PostgreSQL (or MongoDB, MERN option) as the system of record
- Optional Redis cache for exam-session state and rate limiting

3.4 Communications

TLS 1.2+ enforced on all connections; auto-save requests use short-timeout retries with exponential backoff so a brief network blip does not fail the attempt.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow OEP-F-####.

4.1 Authentication & Access Control

Description: Secure registration/login for Admin and Student roles with role-based access control (RBAC) enforced server-side.

Req ID	Requirement (shall...)	Type	Priorty	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OEP-F-001	The system shall allow Admin and Student users to log in with email + password, issuing a role-scoped session/JWT token.	Functional	High	All admin/student	AC-OEP-F-001: Valid credentials return a token scoped to the correct role. Test: TC-Auth-01	Passwords hashed (see OEP-SR-001)
OEP-F-002	The system shall lock a login after 5 consecutive failed attempts for 15 minutes and log the event.	Functional	High	Security	AC-OEP-F-002: 6th attempt is rejected regardless of correctness. Test: TC-Auth-02	

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OEP-F-003	The system shall enforce RBAC at the API level so Student tokens cannot access Admin-only endpoints.	Functional	High	Security	AC-OEP-F-003: Student token calling an admin route returns 403. Test: TC-Sec-Auth-01	See OEP-SR-004
OEP-F-004	The system shall allow an Admin to reset a student's password and force re-verification via email.	Functional	Medium	Admin	AC-OEP-F-004: Reset link expires after 1 hour and works once. Test: TC-Auth-03	

4.2 Question Bank Management

Description: Admin builds and maintains a reusable bank of MCQ questions independent of any single exam.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OEP-F-005	The system shall allow an Admin to create/edit/delete MCQ questions with 2-6 options, exactly one marked correct, marks, negative marks, subject, topic and difficulty tags.	Functional	High	Admin/Faculty	AC-OEP-F-005: A question with zero or 2+ correct options is rejected on save. Test: TC-QB-01	
OEP-F-006	The system shall allow an Admin to search/filter questions by subject, topic and difficulty.	Functional	Medium	Admin/Faculty	AC-OEP-F-006: Filter returns only matching questions. Test: TC-QB-02	
OEP-F-007	The system shall never expose a question's correct-option flag to any Student-facing API response.	Functional	High	Security	AC-OEP-F-007: /attempts/{id}/questions response contains no isCorrect field. Test: TC-Sec-01	See OEP-SR-003

4.3 Exam Creation & Scheduling

Description: Admin assembles an exam from the question bank and schedules a fixed attempt window for one or more student batches.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OEP-F-008	The system shall allow an Admin to create an exam with title, duration, total marks, start time, end time and result-visible-from time.	Functional	High	Admin/Faculty	AC-OEP-F-008: Exam saved with all fields validated (end > start). Test: TC-Exam-01	
OEP-F-009	The system shall allow an Admin to attach questions to an exam manually or via random selection from tagged pools.	Functional	High	Admin/Faculty	AC-OEP-F-009: Random selection respects requested count and difficulty mix. Test: TC-Exam-02	Depends on OEP-F-005
OEP-F-010	The system shall allow an Admin to assign a scheduled exam to one or more student batches.	Functional	High	Admin/Faculty	AC-OEP-F-010: Only students in an assigned batch can see the exam. Test: TC-Exam-03	

4.4 Student & Batch Management

Description: Admin maintains the student roster and batch/section groupings used for exam assignment.

Req ID	Requirement (shall...)	Type	Priorty	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OEP-F-011	The system shall allow an Admin to bulk-import students via a CSV/Excel file (name, email, USN, batch).	Functional	Medium	Admin	AC-OEP-F-011: Valid rows create accounts; invalid rows are reported with reasons, not silently dropped. Test: TC-Stu-01	
OEP-F-012	The system shall allow an Admin to create batches/sections and map students to them (many-to-many).	Functional	Medium	Admin	AC-OEP-F-012: A student can belong to more than one batch. Test: TC-Stu-02	

4.5 Exam Attempt Engine

Description: Enforces a fair, server-timed exam experience for the student, resilient to refresh/crash.

Req ID	Requirement (shall...)	Type	Priorty	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OEP-F-013	The system shall show a Student only the exams assigned to their batch(es) and currently within the scheduled window.	Functional	High	Student	AC-OEP-F-013: An exam outside the window is not listed/accessible. Test: TC-Attempt-00	
OEP-F-014	The system shall allow only one active attempt per (exam, student) pair, blocking a second concurrent session.	Functional	High	Security/Academic Integrity	AC-OEP-F-014: Second login for the same exam is rejected while the first attempt is IN_PROGRESS. Test: TC-Attempt-04	See OEP-SR-005
OEP-F-015	The system shall auto-save the student's current answer state to the server every 10 seconds during an attempt.	Functional	High	Student	AC-OEP-F-015: Refreshing mid-exam restores the last auto-saved answers. Test: TC-Attempt-01	
OEP-F-016	The system shall auto-submit an attempt the instant the server-side timer reaches zero, independent of client clock.	Functional	High	Academic Integrity	AC-OEP-F-016: Attempt status becomes AUTO_SUBMITTED at expiry even if the tab is inactive. Test: TC-Attempt-02	

4.6 Evaluation & Results

Description: Automatic, tamper-resistant scoring performed entirely server-side upon submission.

Req ID	Requirement (shall...)	Type	Priorty	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OEP-F-017	The system shall auto-evaluate a submitted attempt by comparing each Response to the stored correct Option and applying marks/negative-marking rules.	Functional	High	Academic/Business	AC-OEP-F-017: Score = sum(correct marks) - sum(incorrect negative marks); unanswered = 0. Test: TC-Eval-01	

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OEP-F-018	The system shall lock all Responses of an attempt from further modification once its status leaves IN PROGRESS.	Functional	High	Security	AC-OEP-F-018: A write attempt to a submitted attempt's responses is rejected. Test: TC-Sec-02	

4.7 Reporting, Proctoring-lite & Notifications

Description: Analytics, integrity signals and notifications built on top of the core attempt/result data.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OEP-F-019	The system shall generate, per exam, a class performance report and a question-wise difficulty report, exportable to CSV/PDF.	Functional	Medium	Admin/Faculty	AC-OEP-F-019: Export file matches on-screen report totals. Test: TC-Rep-01	
OEP-F-020	The system shall log tab-switch/window-blur events per attempt and present a flagged-attempts view to the Admin.	Functional	Medium	Academic Integrity	AC-OEP-F-020: Two tab switches during an attempt create two ProctoringLog rows visible to Admin. Test: TC-Proc-01	
OEP-F-021	The system shall allow an Admin to manually disqualify or re-permit a flagged attempt, recording the action in an audit log.	Functional	Medium	Admin	AC-OEP-F-021: A disqualified attempt is excluded from the result/report until re-permitted. Test: TC-Proc-02	See OEP-SR-006
OEP-F-022	The system shall send an email notification to affected students when an exam is scheduled and when results become visible.	Functional	Low	Student	AC-OEP-F-022: Notification email is queued within 1 minute of the triggering action. Test: TC-Notif-01	Async job

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs OEP-NF-####.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
OEP-NF-001	The exam attempt page shall load within 2 seconds and support at least 200 concurrent students per exam without response-time degradation beyond 3 seconds (95th percentile).	Performance	High	Load test with 200 simulated students. Test: TC-Perf-01
OEP-NF-002	The application tier shall be horizontally scalable and the database indexed on (exam_id, student_id) for attempts and responses.	Scalability	Medium	Design/code review + query plan (EXPLAIN) shows index usage.
OEP-NF-003	The system shall provide 99.5% availability during any published exam window, excluding pre-announced maintenance.	Reliability	High	Uptime monitoring logs during test exam windows.
OEP-NF-004	The system shall auto-save attempt state at most 10 seconds apart so that a client	Reliability	High	Simulate crash mid-attempt; verify recovered state age. Test: TC-Attempt-01

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
	crash loses no more than the last 10 seconds of activity.			
OEP-NF-005	The UI shall be responsive across desktop and mobile-browser viewports and shall meet WCAG 2.1 AA basics (contrast, keyboard navigation) for exam-taking screens.	Usability/Accessibility	Medium	Manual accessibility audit + Lighthouse score ≥ 90 . Test: TC-UX-01
OEP-NF-006	Core modules (evaluation engine, exam engine) shall carry $\geq 70\%$ automated unit-test coverage.	Maintainability	Medium	Coverage report from CI pipeline.

5.1 Security

5.1.1 Security Objectives

1. Ensure the exam question bank and answer keys remain confidential and are never exposed to a Student via any client-accessible channel (API response, browser dev tools, or page source), before or during an exam.
2. Guarantee the integrity and non-repudiation of submitted answers and computed results: once an attempt leaves IN_PROGRESS status, no student and no unauthorized party can alter its Responses, and every Admin override is attributable via an audit log.
3. Protect user credentials and personally identifiable information (name, email, USN) against unauthorized access through hashing, encryption in transit, and role-based access control.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
OEP-SR-001	The system shall store all passwords using a salted one-way hash (bcrypt or Argon2); plaintext passwords shall never be stored or logged.	Security	High	Inspect DB directly; confirm no plaintext. Test: TC-Sec-Pwd-01
OEP-SR-002	All client-server traffic shall be served over HTTPS/TLS 1.2+, with session cookies marked HttpOnly and Secure.	Security	High	TLS scan + cookie attribute inspection.
OEP-SR-003	The correct-option flag and answer key shall never be included in any API response reachable by a Student role; evaluation shall occur only in the server-side Evaluation Engine.	Security	High	Inspect network responses during an attempt. Test: TC-Sec-01
OEP-SR-004	Role-based access control shall be enforced on every API endpoint server-side (not only hidden in the UI).	Security	High	Attempt admin endpoints with a student token; expect 403. Test: TC-Sec-Auth-01
OEP-SR-005	Each exam attempt shall be bound to a single active session; a second login for the same (exam, student) pair while IN_PROGRESS shall be rejected.	Security	High	Concurrent login test. Test: TC-Attempt-04
OEP-SR-006	All Admin actions that alter question content, results, or attempt status (disqualify/re-permit) shall be written to an append-only audit log with actor, timestamp and action.	Security	Medium	Perform an override; verify immutable log entry. Test: TC-Sec-Audit-01
OEP-SR-007	All user-supplied input shall be validated and persisted via parameterized queries/ORM to prevent SQL/NoSQL injection and stored XSS in rendered question text.	Security	High	Run OWASP ZAP / manual injection payloads against search and question-text fields. Test: TC-Sec-Inject-01

6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: all High-priority functional requirements implemented and verified, no critical NFR or security-requirement failures, and the RTM (Section 9) shows all associated test cases passed.

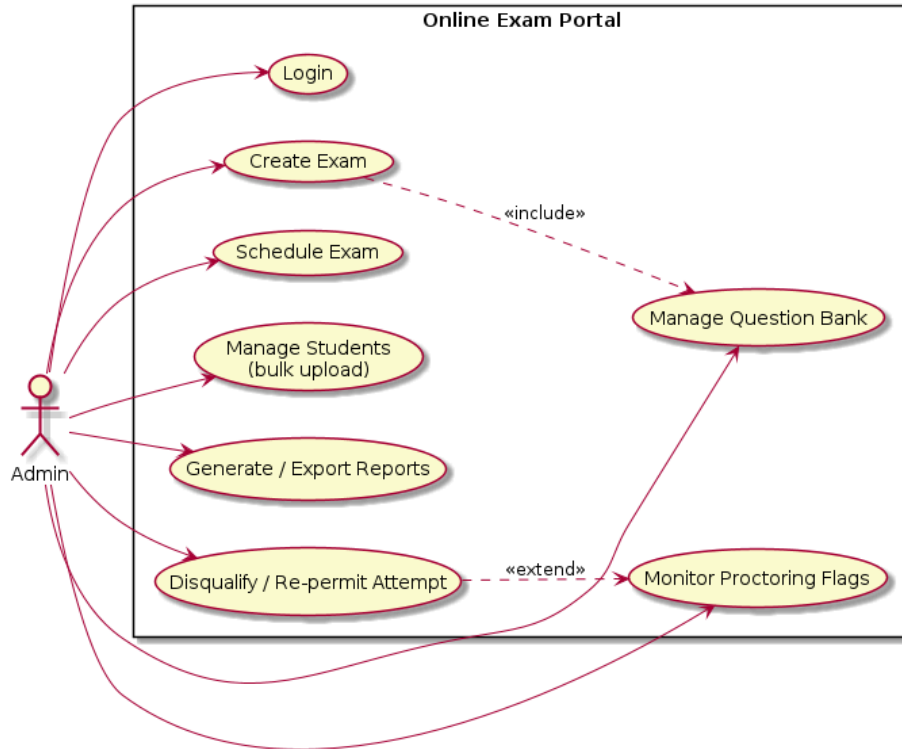
- Acceptance test suites: Authentication & RBAC, Question Bank, Exam Scheduling, Exam Attempt (timer/auto-save/auto-submit), Evaluation, Reporting, Proctoring-lite, Performance, and Security.

7. System Models and Diagrams

This section presents the complete set of UML models and the system architecture diagram for the Online Exam Portal. Each diagram is numbered, titled, and followed by a short explanation of its relevance to the system so that this SRS is sufficient on its own to understand the system's structure and behaviour.

7.1 Use Case Diagram — Admin

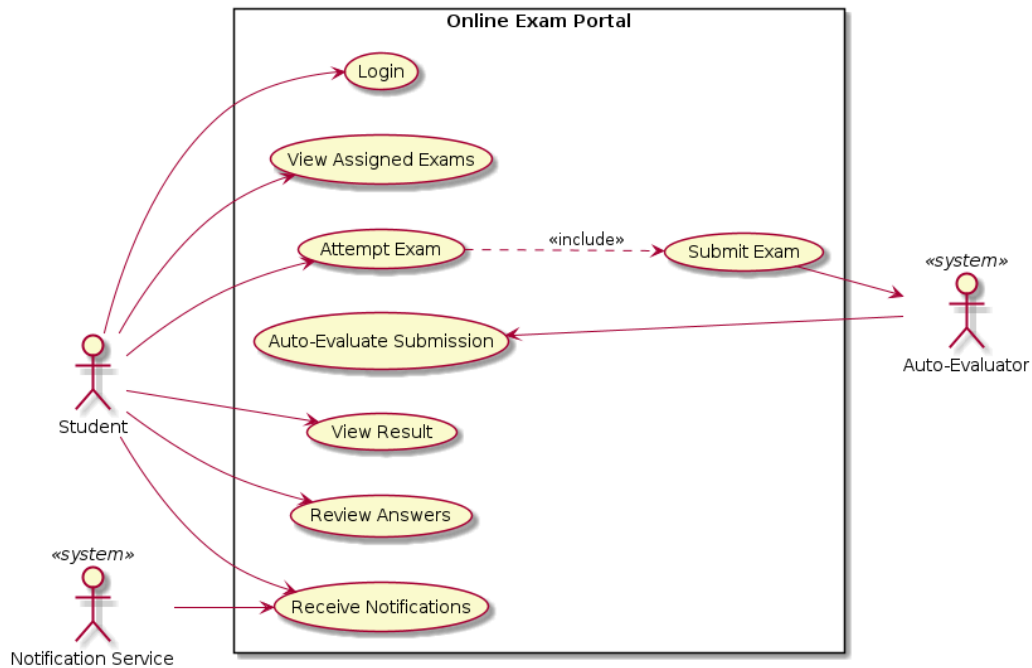
Figure 1: Admin Use Case Diagram



Shows the Admin actor's interactions with the system: logging in, managing the question bank, creating and scheduling exams, managing students, generating reports, and reviewing/acting on proctoring flags. The <<include>> relationship shows that creating an exam always draws on the question bank, and the <<extend>> relationship shows that disqualifying an attempt is an optional extension of monitoring proctoring flags, not a mandatory step.

7.2 Use Case Diagram — Student

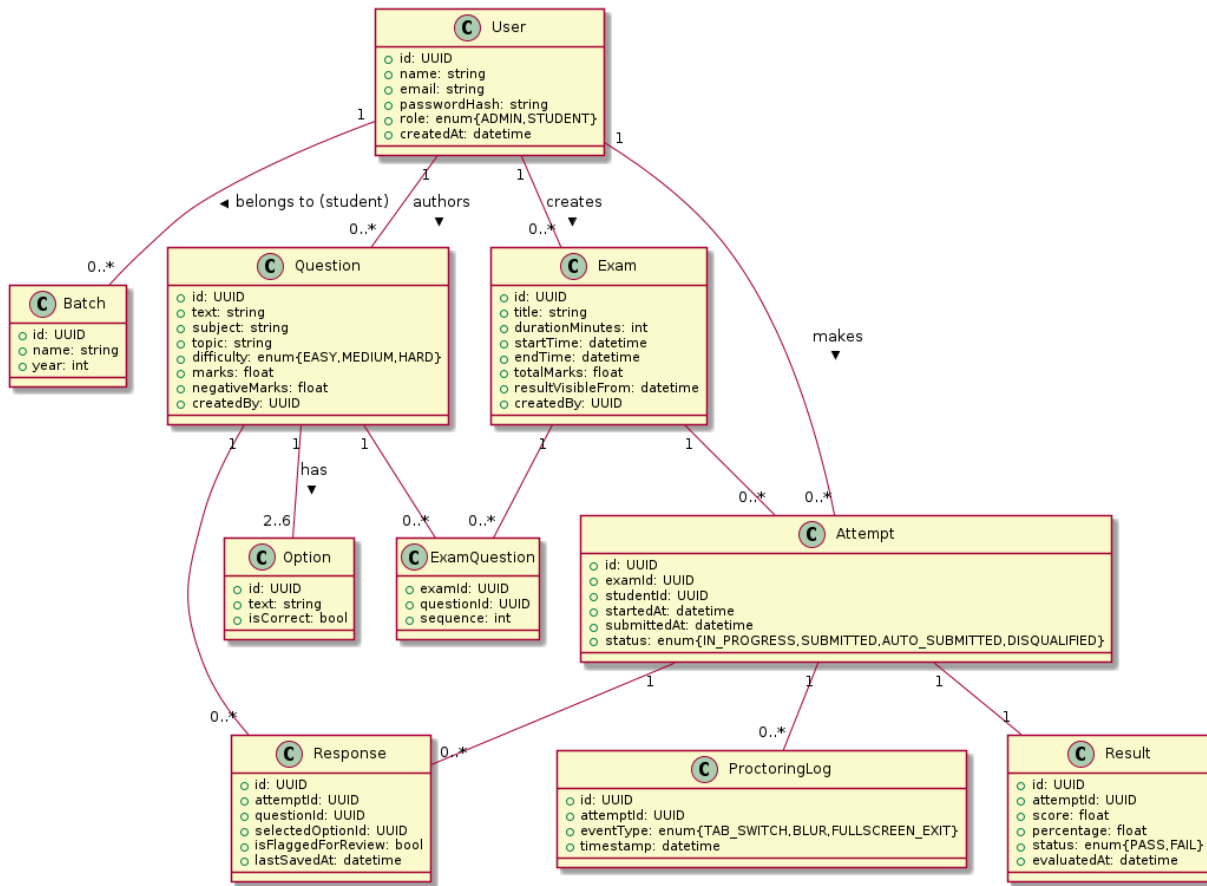
Figure 2: Student Use Case Diagram



Shows the Student actor's journey: logging in, viewing assigned exams, attempting an exam (which includes submitting it), and viewing/reviewing results. Two system actors are modeled explicitly — the Auto-Evaluator, which scores a submission the moment it is submitted, and the Notification Service, which delivers schedule and result alerts — to make clear that evaluation and notification are automated, not manual admin steps.

7.3 Class Diagram

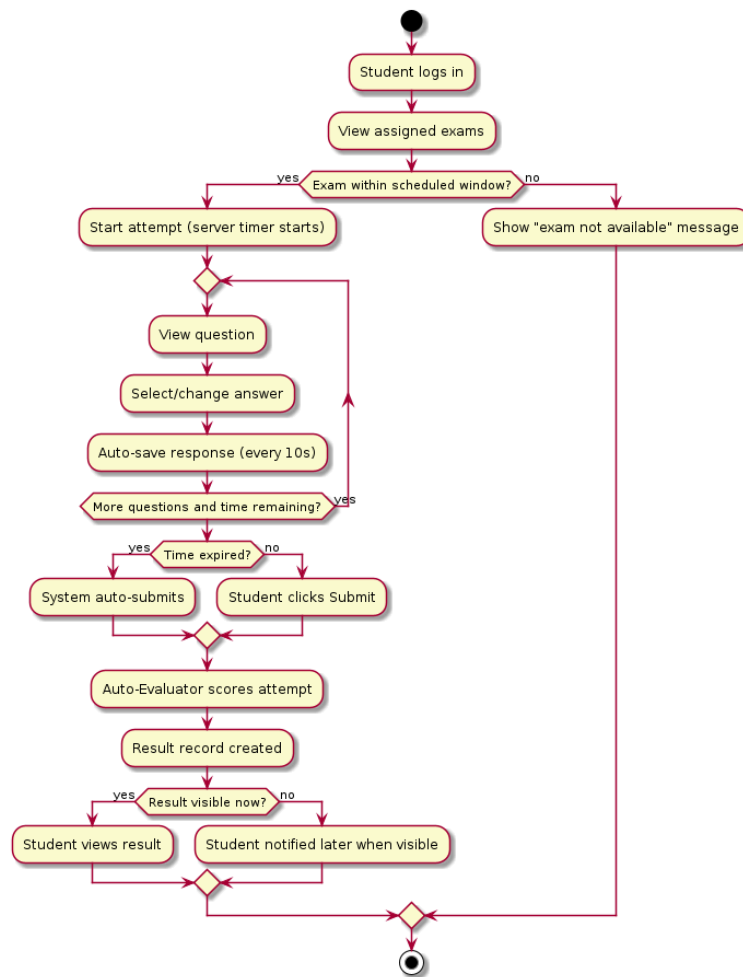
Figure 3: Domain Class Diagram



Models the core domain entities and their relationships: a User (Admin or Student) authors Questions and Exams and makes Attempts; an Exam is composed of Questions through the ExamQuestion association; each Question owns 2–6 Options with exactly one correct; each Attempt owns many Responses (one per question), exactly one Result, and zero or more ProctoringLog entries. This diagram is the direct blueprint for the relational schema in Section 8.

7.4 Activity Diagram

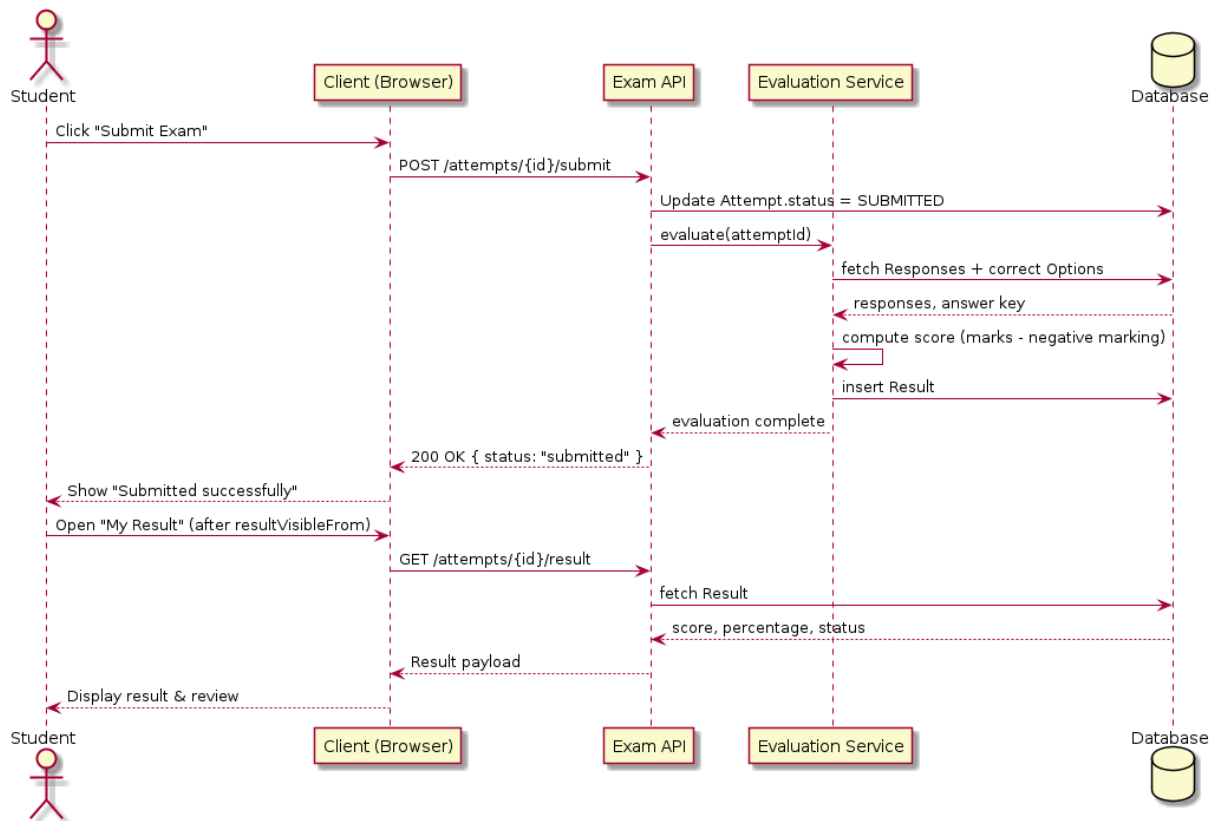
Figure 4: Student Exam-Attempt Activity Diagram



Traces the control flow a Student follows from login to result: the system first checks whether the exam is within its scheduled window; if so, the student answers questions with continuous auto-save until either the timer expires (triggering an automatic submission) or the student submits manually; the Auto-Evaluator then scores the attempt, after which the result is shown immediately or deferred until the admin-configured result-visible time.

7.5 Sequence Diagram

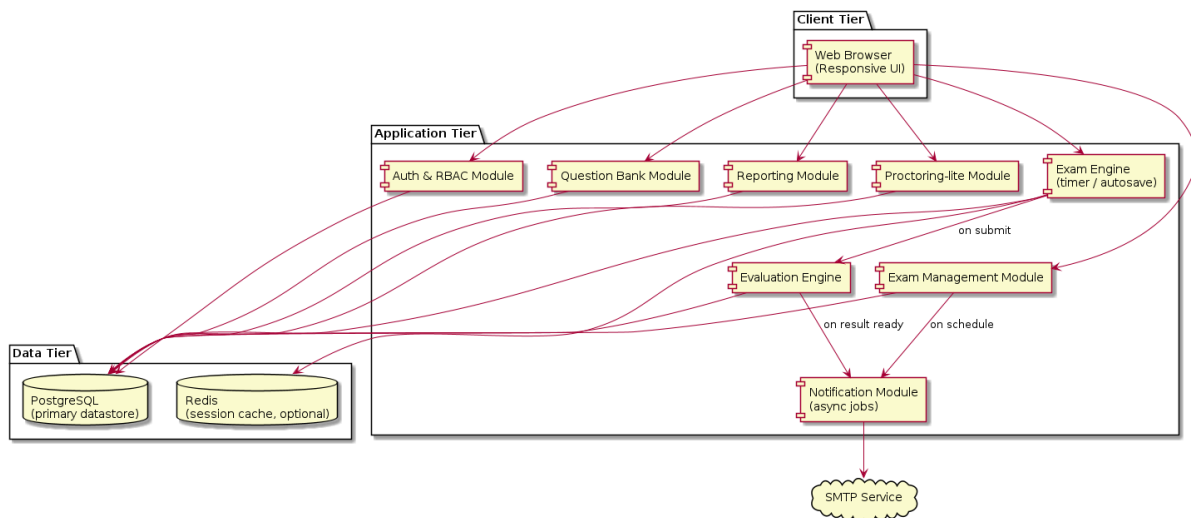
Figure 5: Submit & Auto-Evaluate Sequence Diagram



Details the message flow between the Student, the browser client, the Exam API, the Evaluation Service and the database at the most security-critical moment in the system: submission and scoring. It shows that the answer key is fetched and compared only inside the server-side Evaluation Service — never sent to the client — directly implementing security requirement OEP-SR-003.

7.6 System Architecture Diagram

Figure 6: Three-Tier System Architecture



Depicts the Client Tier (responsive browser UI), Application Tier (Auth & RBAC, Question Bank, Exam Management, Exam Engine, Evaluation Engine, Reporting, Proctoring-lite and Notification modules) and Data Tier (PostgreSQL as the system of record, with an optional Redis cache for session/timer state), plus the external SMTP dependency for notifications. This is the physical/logical realization of the module breakdown referenced throughout Sections 4 and 5.

8. Database Design

The relational schema below directly realizes the Class Diagram in Section 7.3. All tables are normalized to Third Normal Form (3NF), with one documented, deliberate denormalization for read-performance (Section 8.3).

8.1 Entities Overview

Users, Batches, StudentBatches (join), Questions, Options, Exams, ExamQuestions (join), ExamBatches (join), Attempts, Responses, Results, ProctoringLogs. A User with role STUDENT belongs to one or more Batches via StudentBatches; a User with role ADMIN authors many Questions and Exams; an Exam is composed of many Questions through ExamQuestions and is assigned to one or more Batches through ExamBatches; a Student makes one Attempt per Exam (unique on exam_id + student_id), each with many Responses and exactly one Result.

8.2 Table Definitions (Keys & Attributes)

users

Column	Type	Key
id	UUID	PK
name	VARCHAR(100)	
email	VARCHAR(150) UNIQUE	
password_hash	VARCHAR(255)	
role	ENUM('ADMIN','STUDENT')	
usn	VARCHAR(20) NULL	UNIQUE (students)
created_at	TIMESTAMP	

batches

Column	Type	Key
id	UUID	PK
name	VARCHAR(50)	
year	INT	

student_batches

Column	Type	Key
student_id	UUID	PK, FK → users.id
batch_id	UUID	PK, FK → batches.id

questions

Column	Type	Key
id	UUID	PK
text	TEXT	
subject	VARCHAR(50)	
topic	VARCHAR(50)	
difficulty	ENUM('EASY','MEDIUM','HARD')	
default_marks	DECIMAL(5,2)	
negative_marks	DECIMAL(5,2)	
created_by	UUID	FK → users.id
created_at	TIMESTAMP	

options

Column	Type	Key
id	UUID	PK
question_id	UUID	FK → questions.id
text	TEXT	
is_correct	BOOLEAN	

exams

Column	Type	Key
id	UUID	PK
title	VARCHAR(150)	
duration_minutes	INT	
start_time	TIMESTAMP	
end_time	TIMESTAMP	
total_marks	DECIMAL(6,2)	
result_visible_from	TIMESTAMP	
created_by	UUID	FK → users.id

exam_questions

Column	Type	Key
exam_id	UUID	PK, FK → exams.id
question_id	UUID	PK, FK → questions.id
sequence	INT	

exam_batches

Column	Type	Key
exam_id	UUID	PK, FK → exams.id
batch_id	UUID	PK, FK → batches.id

attempts

Column	Type	Key
id	UUID	PK
exam_id	UUID	FK → exams.id
student_id	UUID	FK → users.id
started_at	TIMESTAMP	
submitted_at	TIMESTAMP NULL	
status	ENUM('IN_PROGRESS','SUBMITTED','AUTO_SUBMITTED','DISQUALIFIED')	
(unique)	UNIQUE(exam_id, student_id)	one attempt per student per exam

responses

Column	Type	Key
id	UUID	PK
attempt_id	UUID	FK → attempts.id
question_id	UUID	FK → questions.id
selected_option_id	UUID NULL	FK → options.id
is_flagged	BOOLEAN	
last_saved_at	TIMESTAMP	

results

Column	Type	Key
id	UUID	PK
attempt_id	UUID UNIQUE	FK → attempts.id
score	DECIMAL(6,2)	
percentage	DECIMAL(5,2)	
status	ENUM('PASS','FAIL')	
evaluated_at	TIMESTAMP	

proctoring_logs

Column	Type	Key
id	UUID	PK
attempt_id	UUID	FK → attempts.id
event_type	ENUM('TAB_SWITCH','BLUR','FULL SCREEN_EXIT')	
timestamp	TIMESTAMP	

8.3 Normalization Notes

The schema is in Third Normal Form (3NF): every non-key attribute depends only on its own table's primary key, join tables (student_batches, exam_questions, exam_batches) carry composite keys of exactly the two foreign keys they join with no other dependent data, and there are no transitive dependencies. The one deliberate exception is exams.total_marks, which is a stored, denormalized summary recomputed whenever exam_questions changes — a documented performance trade-off, since total_marks is read on every result view but changes only when an admin edits the question list.

9. Requirements Traceability Matrix (RTM)

Every requirement ID below traces to the section defining it, the implementing module, and its verifying test case(s), confirming full traceability from requirement through design to test.

Req ID	Requirement (short)	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)
OEP-F-001	Login with RBAC	4.1	AuthModule	TC-Auth-01, TC-Sec-Auth-01	N
OEP-F-005	Question bank CRUD	4.2	QuestionBankModule	TC-QB-01	N
OEP-F-007	Hide answer key from student API	4.2 / OEP-SR-003	QuestionBankModule / ExamEngine	TC-Sec-01	N
OEP-F-009	Build exam from question bank	4.3	ExamManagementModule	TC-Exam-02	N
OEP-F-014	Single active attempt per exam	4.5 / OEP-SR-005	ExamEngine	TC-Attempt-04	N
OEP-F-015	Auto-save every 10s	4.5	ExamEngine	TC-Attempt-01	N
OEP-F-016	Server-side auto-submit on timeout	4.5	ExamEngine	TC-Attempt-02	N
OEP-F-017	Auto-evaluate with negative marking	4.6	EvaluationEngine	TC-Eval-01	N
OEP-F-019	Class & question-wise reports	4.7	ReportingModule	TC-Rep-01	N
OEP-F-020	Tab-switch proctoring log	4.7	ProctoringModule	TC-Proc-01	N
OEP-NF-001	Response time / concurrency target	5	ExamEngine / Infra	TC-Perf-01	N
OEP-NF-003	99.5% availability during exam windows	5	Infra / Ops	Ops monitoring	N
OEP-SR-001	Password hashing	5.1.2	AuthModule	TC-Sec-Pwd-01	N
OEP-SR-003	Answer key never sent to client	5.1.2	ExamEngine	TC-Sec-01	N
OEP-SR-005	Single-session-per-attempt enforcement	5.1.2	ExamEngine	TC-Attempt-04	N